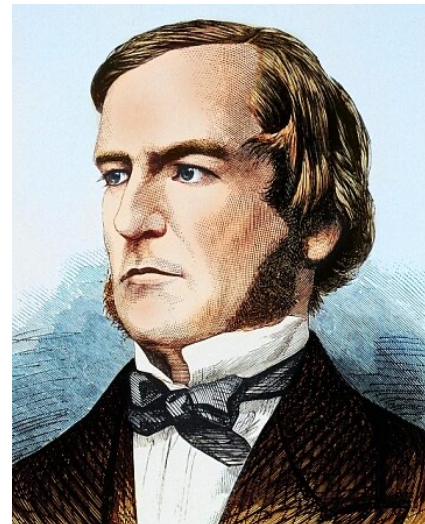


Inductive Reasoning

Logical Reasoning and Human Nature

- Historically, many researchers believed that logical reasoning is an essential part of human nature
 - Aristotle
 - Boole (1854). A book on logical calculus
 - “An investigation of the laws of thought”
- Rational behavior = logical thinking (deductive reasoning)



Inductive vs. Deductive Reasoning

- **Deductive reasoning:**
 - Conclusion follows logically from premises
- **Inductive reasoning:**
 - Conclusion is likely based on premises
 - Involves a degree of uncertainty
- Reasoning in real-world is often based on induction
- In computer science:
 - Logical reasoning is often associated with symbolic AI
 - Inductive reasoning is often associated with statistical machine learning

Deductively Valid?

- Premise: All cars have wheels
 - Premise: All wheels are round
 - Conclusion: All cars have round wheels
-
- Premise: I have a diamond
 - Premise: Most diamonds are shiny
 - Conclusion: My diamond is shiny
-
- Premise: John is 93
 - Conclusion: John will not do a double back flip today

Inductive Reasoning

- Reason from **observable** information to **unobservable** and **uncertain** information

What makes people smart?

- Memory? No.
- Deductive inference? No.
- Intuitions and inductions.

Everyday Inductive Leaps

- How can people learn so much about the world from such limited evidence?
 - Learning concepts from examples – “horse”



Real World Inductive Inferences

- Medical diagnosis:

- Symptoms, test outcomes (observable) → Diseases (unobservable)

- Scientific reasoning:

- Experimental data (observable) → Hypotheses (unobservable)

- Law:

- Facts (证据) (observable) → Guilt (罪行) (unobservable, uncertain)

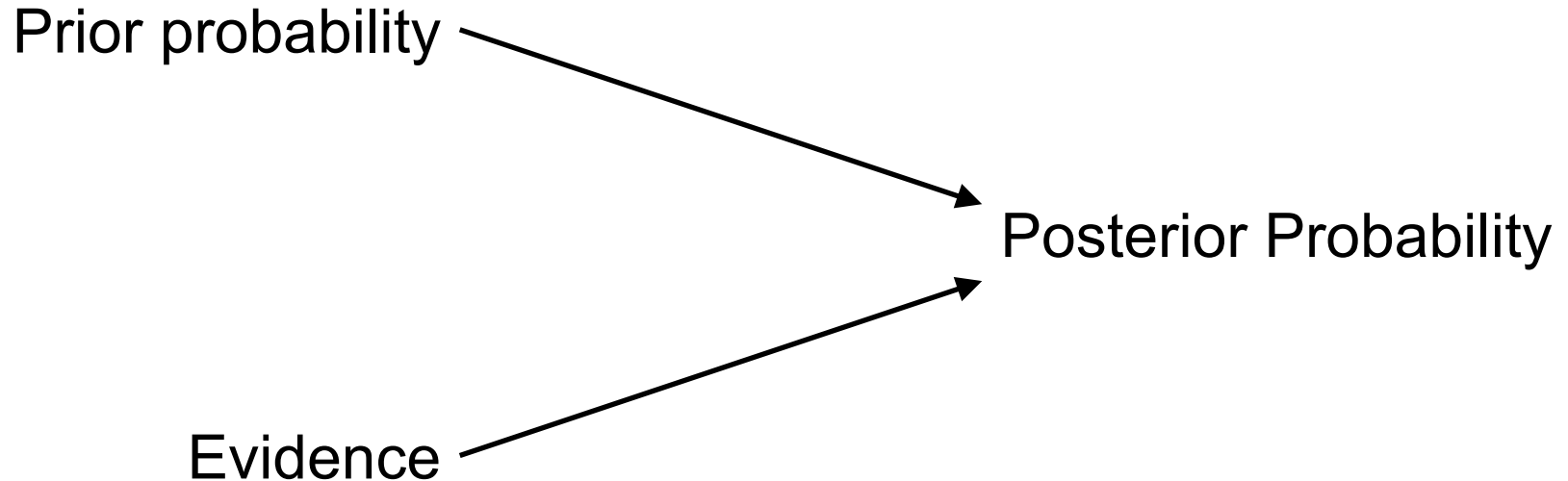
Reasoning under Uncertainty

- **Bayes rule** tells us how to optimally reason with uncertainty
 - Bayes estimation
- Allows us to say how we believe something to be true based on **prior beliefs** and **new available evidence**



Thomas Bayes (1702-1761)

Bayes Rule



Bayes rule tells us how the available evidence should alter our belief in something being true

Do People Reason like Bayes Rule?

- Problems understanding **conditional probability**
 - Doctors need to calculate the probability of disease given the observed symptoms: $P(\text{disease} \mid \text{symptoms})$
 - Sometimes $P(\text{symptoms} \mid \text{disease})$ is used incorrectly when reasoning about the likelihood of a disease
- Why is this wrong?

The Base Rate is Important

- To get $P(\text{disease} \mid \text{symptom})$, you need to know about $P(\text{symptom} \mid \text{disease})$ and also the **base rate** -- prevalence of the disease before you have seen patient
- More intuitive example:
 - What is the probability of being tall given you are player in the NBA?
 - What is the probability of being a player in the NBA given that you are tall?

$$P(\text{NBA player} \mid \text{tall}) \neq P(\text{tall} \mid \text{NBA player})$$

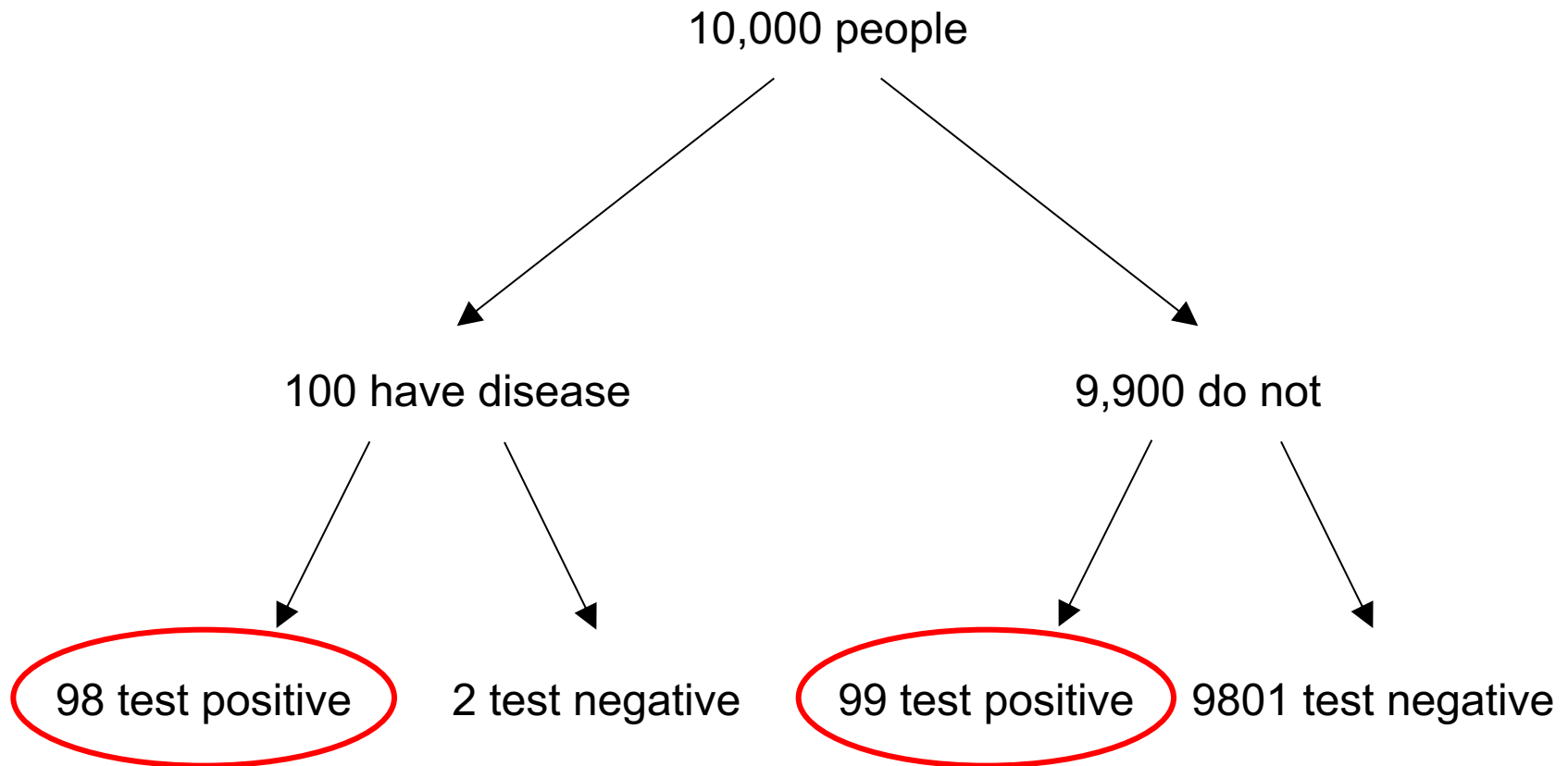
Reasoning with Base Rates

- Suppose there is a disease that affects 1 out of 100 people
- There is a diagnostic test with the following properties:
 - If the person has the disease, the test will be positive 98% of the time
 - If the person does not have the disease, the test will be positive 1% of the time
- A person tests positive, what is the probability that this person has the disease?
 - Frequent answer =
 - Correct answer $\approx$

Are we really that bad in judging probabilities?

- According to some researchers (e.g., Gigerenzer), it matters **how** the information is presented and processed
- Processing **frequencies** is more intuitive than **probabilities**

A Counting Heuristic (in tree form)



$$P(\text{disease} \mid \text{test positive}) = 98 / (98 + 99) \approx .50$$

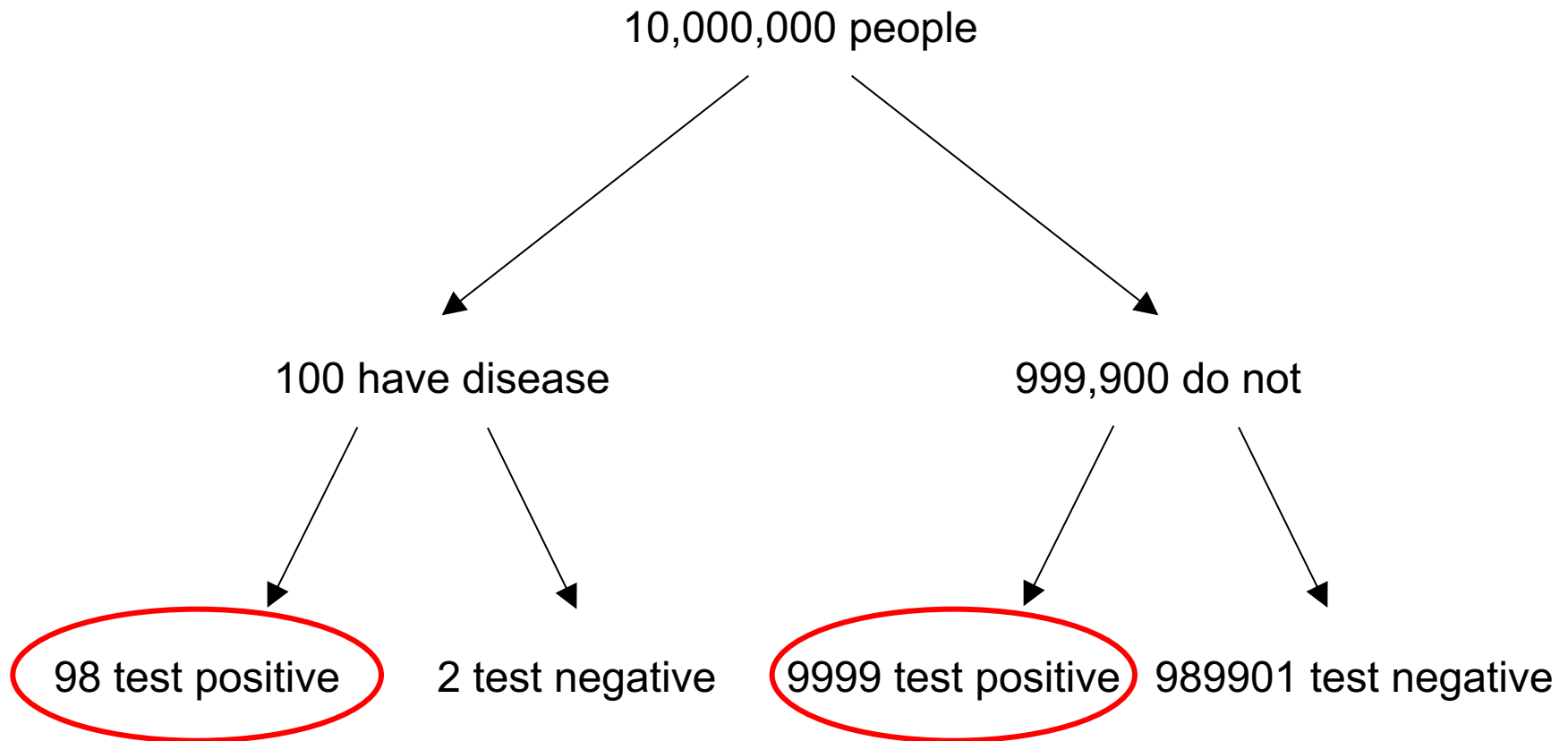
The Same Thing in Words ...

- Let's take 10,000 people
- On average, 100 out of 10,000 actually have the disease and 98 of those will test positive (98% true positive rate)
- Among the 9,900 who do not have the disease, the test will falsely identify 1% as having it. $1\% \text{ of } 9,900 = 99$
- On average, out of 10,000 people:
98 test positive and they have the disease
99 test positive and they do not have the disease
- Therefore, a positive test outcome implies a $98/(98+99) \approx 50\%$ chance of having the disease

Change the Example

- What now if the disease affects only 1 out of 10,000 people?
- Assume same diagnosticity of test
(98% true positive rate, 1% false positive rate)
- A person tests positive, what now is the probability that this person has the disease?

A Counting Heuristic (in tree form)



$$P(\text{disease} \mid \text{test positive}) = 98 / (98 + 9999) = .0097$$

(smaller than 1%)

Bayes Rule

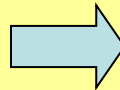
- The previous example essentially is a simple way to apply Bayes rule:

$$P(disease | positive) = \frac{P(positive | disease)P(disease)}{P(positive | disease)P(disease) + P(positive | not\ disease)P(not\ disease)}$$

$$P(positive | disease) = .98$$

$$P(positive | not\ disease) = .01$$

$$P(disease) = .0001$$



$$P(disease | positive) = .0097$$

Normative Model

- Bayes rule tells you how you should reason with probabilities -- it is a prescriptive (i.e., **normative** (标准模型)) model
- But do people reason like Bayes?

In certain circumstances, we observe **base rate neglect** (基率忽视)

The **base rate fallacy**, also called **base rate neglect** or **base rate bias**, is an error that occurs when the conditional probability of some hypothesis H given some evidence E is assessed without taking into account the prior probability ("base rate") of H and the total probability of evidence E